

KYAMBOGO UNIVERSITY.

THE DETERMINANTS OF HOUSEHOLD CONSUMPTION EXPENDITURE IN UGANDA.

BY

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20/U/BEK/2091/PD

A RESEARCH PROPOSAL SUBMITTED TO THE DEPARTMENT OF ECONOMICS AS A  
PARTIAL FULFILLMENT FOR THE AWARD OF A DEGREE OF BACHELORS OF  
ECONOMICS AND STATISTICS (BES) OF KYAMBOGO UNIVERSITY.

AUGUST, 2024.

## **DECLARATION**

**I NAMUGALU HILLARY MILDRINE** hereby declare that this report entitled “*Determinants of Household Consumption Expenditure* “ is my original work and has not been presented by anyone for the award of a degree in any other university or institution of higher learning.

Signature ..... Date .....

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### **APPROVAL**

This research report is submitted for the award of Bachelor of Economics and Statistics of Kyambogo University with approval from the University academic supervisor.

SIGNATURE: ..... DATE: .....

**Mr. Semanda Edward Patrick.**

## **DEDICATION**

I hereby dedicate this report to whole heartedly to my family for the support rendered during the period of carrying out research.

## **ACKNOWLEDGEMENT**

My sincere gratitude goes to Mr. Semanda Edward Patrick for the dedicated efforts in supervision and mentorship during the time of this research work that has enabled the completion of this quality research work.

I also extend my sincere appreciation to my mom, Ms. Nasaali Grace for the tireless efforts for the financing throughout the entire period of education and conducting research.

Appreciation to my friends Mr. Dongo Charles-senior policy analyst at the ministry of justice and constitutional affairs for mentoring me and for the encouraging words during hard times while carrying out the research.

May the Almighty God bless you all for your tireless efforts and sacrifices rendered in your respective capacities.

## LIST OF ABBREVIATIONS/ACRONYMS

|               |                                                   |
|---------------|---------------------------------------------------|
| <b>GDP</b>    | <b>Gross Domestic Product</b>                     |
| <b>OLS</b>    | <b>Ordinary Least Squares</b>                     |
| <b>NPISHs</b> | <b>Non-Profit Institutions Serving Households</b> |
| <b>WEO</b>    | <b>World Economic Outlook.</b>                    |
| <b>UBOS</b>   | <b>Uganda Bureau of Statistics</b>                |
| <b>VAR</b>    | <b>Vector Auto regression</b>                     |
| <b>ADF</b>    | <b>Augmented Dickey Fuller</b>                    |
| <b>WDI</b>    | <b>World Development Indicators</b>               |

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## **ABSTRACT**

This research report examines the Determinants of Households Consumption Expenditure. The study aims to understand the factors that shape households spending behavior and shed light on the macroeconomic variables that impact private consumption.

Using econometric approach, the study analyzed a comprehensive dataset encompassing relevant economic indicators, including household's consumption expenditure, gross savings, public debt and expenditure. The data cover a specific time period of time series data from 1990-2020 from World Bank Development Indicator (WDI), allowing for various analysis like the Augmented Dickey-Fuller (ADF) test for unit root that shows whether there is stationarity basing on the hypothesis set, testing for normality using sktest of the residuals to show whether the variables are normally distributed, test for serial correlation using Breusch- Godfrey, test for constant variance.

The findings of the study indicate that gross savings, public debt and expenditure have a negative and significant effect on household consumption expenditure. The research report concludes that policies aimed at managing gross savings and expenditure can positively influence household's consumption behavior.

## **CHAPTER ONE: INTRODUCTION**

### **1.1. Background of the Study.**

Consumption expenditure refers to the total value of all goods and services consumed by households. It includes spending on necessities such as food, clothing, housing, healthcare, education and other services. This type of expenditure is a significant component of gross domestic product (GDP) and is often used to gauge the economic health of a country, as higher consumption typically indicates stronger economic activity and confidence among consumers.

According to Castilla & walker (2015), consumption is a useful indicator since it involves the total money spent on final goods and services by individuals and households for personal use and enjoyment in the economy. Contemporary measures of consumption expenditure include all private purchases of durable goods, non-durable goods such as household washing machines, television etc. Household consumption expenditure exclude purchases of residences but includes owner-occupied residences imputed rent. However, consumption expenditure involves spending of the household incomes on all domestic costs (by residents and non-residents) for individual's needs, among other things includes expenditures on goods and services, household consumption expenditure varies substantially the population due to incomes of households, economic structure and the degree of urbanization (Morton 2019). It makes two-thirds of GDP, so fluctuations in consumption are key element of booms and recessions of business cycle. The consumption decision is also crucial for long-run analysis because of its role in economic growth. Household consumption expenditures, investment, public expenditures and net export are the components of GDP. Due to the high share in GDP, consumption expenditure are taken into account in macroeconomics policies for fiscal planning. Policy makers try to predict how consumers will behave in the face of income fluctuations (4). Household's decision on how much to consume or save is a microeconomic question as it deals with the individual units of economy. However, it has an influence on the economy as a whole since aggregate household consumption determines the behaviour of the economy in the short and long run. The effects of changes in consumer spending on aggregate demand are crucial for short run analysis. Consumption forms about two thirds of GDP; hence fluctuation in the level of consumption can create shocks in the economy. In the short run, the marginal propensity to consume has a hand in determining the fiscal policy multiplier. Household consumption also has a part to play in long run analysis because of its influence on the growth of the economy. Consumption is the amount of service of a good that is used up in any one

period while consumption expenditure is the expenditure on consumer goods in a period. The consumption function is used to calculate the total consumer spending in an economy. Every now and then individual households and the government come across the status of consumption. Despite the fact that consumption is one of the fundamental determinants of aggregate economic activities there is no consensus among economists about the consumption hypothesis which represents consumer behaviour. Various schools of thought have propounded ways of measuring consumption. Of all the schools of thought, the most commonly used is the permanent income hypothesis.

***Table 1 Uganda economic indicators***

| <b>Indicator</b>                           | <b>Period</b>       | <b>Value (Trillion UGX)</b> | <b>Value (USD) (Billion)</b> |
|--------------------------------------------|---------------------|-----------------------------|------------------------------|
| Consumption expenditure                    | September 2015-2023 | 3.52                        | \$32.32                      |
| Government consumption expenditure         | September 2015-2023 | 3.52-4.72                   | \$32.32-\$43.45              |
| Household final consumption expenditure.   | 2022                | 47.26                       | \$32.32                      |
| GDP at current prices                      | 2022                | 122.32                      | \$105.67                     |
| Annual growth rate (household expenditure) | 2022-2006           | 14.82                       | N/A                          |
| Annual growth rate (GDP)                   | 2002-2006           | 8.27                        | N/A                          |

***Source: Uganda's General Government Total Expenditure and Uganda-Final Consumption expenditure, 1980-2021.***

Let's use income and some forms of consumption expenditure such as healthcare expenditure and education expenditure to track the trends:

In 2023, Uganda had the lowest average gross income in the Middle East and Africa but saw a 6.2% rise in per capita disposable income due to robust economic performance. Despite this, income inequality remained significant, with the top 10% households earning 33.1% of total disposable income. Notably, during the first quarter of FY24, Uganda's GDP grew by 5.3%, supported by an oil-related construction boom and robust growth in agriculture, private investments and employment growth also reinforced domestic demand.

***Table 2 Uganda's macroeconomic performance and income-related indicators from 2018-2023.***

| <b>YEAR</b> | <b>GDP(current US\$)</b> | <b>GDP per capita</b> | <b>GDP growth (annual %)</b> | <b>Inflation (consumer prices, annual %)</b> | <b>Fiscal Deficit</b> |
|-------------|--------------------------|-----------------------|------------------------------|----------------------------------------------|-----------------------|
| 2018        | \$38.28billion           | \$797.5               | 5.6%                         | 2.1%                                         | -7.1%                 |
| 2019        | \$40.86billion           | \$849.3               | 7.6%                         | 2.8%                                         | -9.0%                 |
| 2020        | \$39.99billion           | \$828.9               | -1.2%                        | 2.2%                                         | -7.4%                 |
| 2021        | \$42.47billion           | \$880.7               | 5.6%                         | 7.2%                                         | -5.3%                 |
| 2022        | \$45.57billion           | \$947.4               | 5.8%                         | N/A                                          | N/A                   |
| 2023        | N/A                      | N/A                   | 6.5%                         | 6.5%                                         | -4.6%                 |

Uganda's commitment to education has seen fluctuations over the years. In 2010, education received 8.90% of the government budget, and this percentage remained relatively stable until 2015. However, from 2016-2019 there were variations reaching a peak of 8.90% in 2018 and declining to 8.21% in 2019. The recent trend shows a slight increase with 8.48% in 2022. While Uganda faces challenges in poverty reduction and education, sustained investment in education remains crucial for human capital development.

According to the World Health Organisation (WHO) Global Health Expenditure database, Uganda's current health expenditure as a percentage of GDP was approximately 3.96% in 2020. Uganda's investment in health per capita (PPP-adjusted) stands at \$6.8, which is one of the lowest

in the region and reflects the average health spending per person in the country. Currently, households and external development partners finance a combined 84% of Uganda's total health spending hence government funding remains relatively low. (8)

These trends suggest that Uganda's consumption expenditure is on a downward trajectory, with both government and household spending contributing to the country's economic growth. Projections indicate continued growth in income but low consumption expenditure, reflecting the country's economic under development.

### **1.2 Problem Statement.**

This research report addresses the need for a better understanding of the determinants of household's consumption expenditure. Specifically, the study aims to examine the effect of gross savings, public debt and government expenditure on household's consumption expenditure in Uganda. While previous research has examined some of these factors individually, there was a need to understand the joint effect of these factors on Households Consumption Expenditure. The existing literature has produced mixed results on the significance of the determinants. Therefore, further research is needed to provide a clear understanding of these relationships and informed decisions aimed at promoting economic growth and development

### **1.3. Objective of the study.**

The overall objective of the study is the determinants of household consumption expenditure in Uganda while the specific objectives of the study are;

1. To determine the effect of public debt on household consumption expenditure in Uganda.
2. To find out the effect of domestic savings on household consumption expenditure in Uganda.
3. To find out the effect of government expenditure on household consumption expenditure in Uganda.

### **1.4. Study hypothesis.**

- (i) Public debt has no significant effect on household consumption expenditure in Uganda.
- (ii) Domestic savings have no significant effect on the household consumption expenditure in Uganda.

(iii)Government expenditure has no significant effect on the household consumption expenditure in Uganda.

### **1.5. Scope of the study.**

This study focuses on Uganda that is located in the eastern part of Africa bordered by Tanzania in the south, to the north by South Sudan, to the west by DRC, to the east by Kenya and to the south west by Rwanda. This study will cover thirty years (1990-2020). This study also looks at GDP, savings, inflation and government spending as the determinants of household consumption expenditure.

### **1.6. Significance of the Study.**

The study provides information and framework to the citizens in Uganda about consumption expenditure patterns. By analyzing these gaps, we can inform nutrition sensitive interventions, policy making and sustainable food security strategies in Uganda.

## **CHAPTER TWO: LITERATURE REVIEW**

### **2.0. Introduction**

This chapter presents the literature reviewed on the different objectives of the study. It consists of both theoretical and empirical reviews from previous studies.

### **2.1. Theoretical review**

There are various theories (10) that were used to explain consumption behaviors and these include; behavioral economics, classic economic theory, Keynesian theory, marginal utility theory, Veblenian theory of conspicuous consumption, permanent income hypothesis and life cycle hypothesis but we will focus on the following theories in relation to consumption expenditure;

#### **2.1.1 Relative Income Hypothesis (RIH).**

American economist James Duesenberry's relative income hypothesis was first proposed in the income saving and theory of consumer behavior in 1949 (2). It states that an individual's attitude toward consumption and saving is influenced more by their income than others. The level of consumption obtained in a previous time also influences current absolute and relative income levels i.e. if a family attains a certain level of consumption; they find it difficult to lower it. This challenged Keynes' notion of consumption behavior by introducing psychological factors associated with habit formation and social interdependencies based on relative income concerns.

According to Duesenberry (2), the intensity of any individual's desire to increase his consumption expenditure is determined by ratio of his expenditure to some weighted average of the expenditures of others with whom he interacts. This theory has four important components which include the following;

Individuals are more concerned with their relative wellbeing than their absolute well-being.

Poor people spend greater portion of their income than the rich people to narrow the consumption gap.

Current absolute and relative income levels and previous consumption levels determine present consumption.

Consumption within a family is influenced by its income compared to other families.

According to the absolute income hypothesis, an individual allocates disposable income between spending and saving while in the relative income hypothesis, current income has little impact on consumption. People spend in line with their relative income situation.

### **2.1.2 Permanent Income Hypothesis (PIH).**

The theory states that consumers spend their earnings at a level in accord with their estimated future income over the long term. Individuals view this expected income level as their permanent income level which they believe is safe to spend. The theory was developed by Milton Friedman and published in 1957(9). According to him, individual expectations influence consumer spending and the expectations vary depending on how consumers view the factors that impact future income. Hence, any policy decision affecting consumers' perception of their long-term income will increase their spending.

In addition, an individual's future earnings expectations also depend on their liquidity. One who does not possess any asset may already have a habit of spending money regardless of their current or future earnings. Therefore, changes in permanent income can occur via incremental salary increases or the assumption of better long-term jobs with improved and sustainable pay. If workers' expectations increase, they may spend more.

The permanent income hypothesis of consumption offers a framework to understand how households might react to the variations in earnings while making consumption expenditure decisions in the near term. (7)

### **2.1.3 Life-Cycle Hypothesis (LCH).**

The concept was developed by economists Franco Modigliani and his student Richard Brumberg in the early 1950s (11). It assumes that people plan their spending throughout their lifetimes, factoring in their future incomes. According to the concept, they take on debt when they are young, assuming future income will enable them to pay it off. They then save during middle age in order to maintain their level of consumption when they retire. The theory has assumptions which include;

The theory assumes that people deplete their wealth during old age. Often, however, the wealth is passed on to children or older people may be unwilling to spend their wealth.

The theory assumes that people earn most when they are of working age. However some people choose to work less when they are relatively young and continue working part time when in their retirement age.

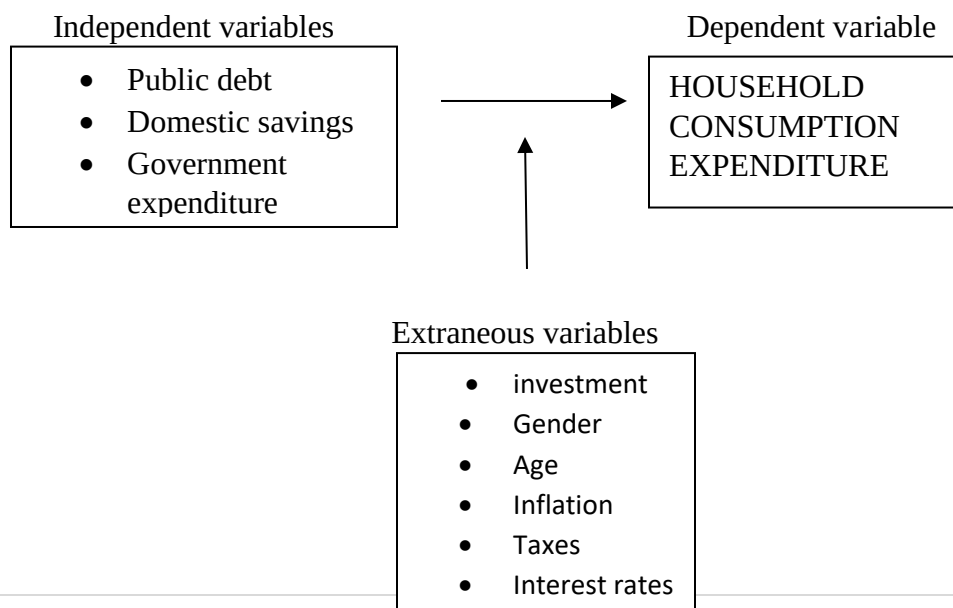
It also assumes that people plan ahead when it comes to building wealth but many procrastinate or lack the discipline to save. The graph of the LCH shows a hump shaped pattern of wealth accumulation that is low during youth and old age and high in the middle age.

As a result, one implication is that younger people have a greater capacity to take investment risks than older individuals who need to draw down accumulated savings which remains a widely accepted tenet of personal finance.

The LCH replaced the Keynesian theory developed by John Maynard Keynes in 1937. Keynes believed that savings were just another good and that the percentage that individuals allocated to their savings would grow as their incomes rose and this presented a potential problem in that it implied that as a nation's incomes grew a savings glut would result and aggregate demand and output would stagnate. Hence relevancies to the theory such as Keynes' theory did not address people's consumption patterns over time for example a middle-aged person as a head of a family will consume more than a retiree. (13)

## 2.2. Conceptual framework

**Figure 1** conceptual frame work showing relationship between the variables both independent and dependent



### **2.3 Household's consumption expenditure pattern and its determinants.**

Individual consumers are assumed to be in the best position to judge their own needs and preferences to make their own choices. It's unbiased to assume that people know what they are looking for and have reasons for their preferences when they choose one consumption pattern over another. Yet millions of people face a narrow range of consumptions, which prevents them from enlarging their capabilities. They may not be able to get enough food, may lack health care services or may have little access to transport beyond their own feet. There are many factors causing these constraints on consumption option of different individuals. Income is not the only one but other factors include government expenditures, household saving status, interest rates education status etc.

#### **2.3.1 The effect of household saving status on household consumption expenditures.**

The author in (12) defines household savings as the difference between a household's disposable income and its consumption. The household savings rate indicates that a household spends more than it earns as regular income and finances some of its expenditure through credit or through gains from the sale of assets or by running down cash reserves. Generally, saving may be thought of as resources created or outputs produced in the current period that are not consumed in the current period but rather are made available for future consumption. With this idea in mind, savings is alternatively defined as income minus consumption, or the change in wealth, or the supply of capital. Households' savings are an important component of individuals and society's well-being. Households save owing to the following reasons; to build a reserve against unforeseen contingencies (precautionary motive), for smoothing consumption at different stages in life cycle due to income fluctuations (life cycle motive), to enjoy interest and appreciation (inter temporal substitution motive), and to enjoy a gradually increasing expenditure (improvement motive). In addition, they save to enjoy a sense of financial freedom and independence (independence motive), to secure finance to carryout speculative or business project (enterprise motive), to bequeath a fortune (bequest motive) (3).

#### **2.3.2 Consumption and income.**

Income gives people have the ability to buy nutritious foods, to pay for health care and education for their families, to pay for water from a tap instead of walking for many hours to collect it from

a well. The increasing dependence of much consumption on private income means that changes in income have a dominant influence on changes in consumption. When income rises steadily consumption rises for most of the population. But for some reason when income declines, consumption also falls sharply, with devastating consequences for human wellbeing. It seems reasonable to expect that consumption spending by households will be closely related to their disposable income, which equals the income households receive less taxes they pay. The relationship between consumption and personal disposable income is called consumption function. Consumption depends on real disposable income, wealth, the overall price level, expectations etc. This means that the decision to spend income on consumption goods largely is determined by these factors. Some of these factors have a positive impact on consumption expenditure, others have a negative impact. Of all these factors the most important is the level of real disposable income. If the real disposable income increases, the individual and households are likely to increase their consumption spending. Therefore, there is a positive relationship between real disposable income and consumption.

## **2.4. Related literature**

The empirical analysis of the household consumption behavior was pioneered by the work of Engel. In the mid-19th century, he proposed his famous law of consumption, which is formulated based on the family budget studies of different countries. The establishment of this law itself is thus taken as the first significant quantitative approach contributed to the field. The empirical evidence explained below indicates that the determinants of the household consumption in many areas are similar but, some socio-economic factor, shows different effect based on specific conditions of the study area. Even though the previous studies are so good, they are not taken into account some determinants like disposable income, saving and some other socio-economic factor that determine household consumption pattern.

### **1. Savings**

According to John Maynard Keynes in the absolute income hypothesis (3), consumption behavior depends on current disposable income after tax payment. It is also known as Keynes' consumption function as it explains the relationship between income and consumption.

Keynes asserted that when income increases, consumption expenditure also increases but at a diminishing rate. Therefore, as people earn more, they tend to spend more but not necessarily proportionally.

The absolute income hypothesis is expressed as:

$$C_t = a + bY_t$$

$C_t$ - Consumption at time (t)

a-Autonomous consumption/ constant

b-Marginal propensity to consume

$Y_t$ - Disposable income at time (t)

While successful for short term modeling, applying this model over time frames has proven less effective hence economists have explored other consumption models due to various limitations.

## 2. Public debt

Keynesian economists including Abba Lerner believe that government borrowing even for unproductive projects can stimulate economic output. This occurs by absorbing excess savings and boosting aggregate demand, especially during recessions.

Keynes saw public debt as a means to mobilize saving, encourage investment and improve living standards. Governments borrow money when their revenue falls short of expenditures using debt as an instrument to manage economic stability.

Keynes emphasized that increasing public debt does not necessarily raise interest rates but successful policy management can mitigate upward pressure on interest rates, particularly concerning long-term rates and the term structure of rates.

## 3. Government expenditure

Keynesian economists emphasize the role of aggregate demand (total spending in the economy) in influencing economic activities. During economic downturns (recessions or depressions), Keynesian's advocate for increased government spending to stimulate demand. Governments should use fiscal policy to stabilize the economy. For example;

- i. Expansionary fiscal policy: during recessions, government should increase its spending and reduce taxes to boost demand.
- ii. Contractionary fiscal policy: during inflationary periods, the government should decrease on its spending and raise taxes to control demand.

Keynesians believe that government spending has a multiplier effect that is each dollar spent generates additional income and spending in the economy.

## CHAPTER THREE: RESEARCH METHODOLOGY

### 3.0. Introduction

This chapter consists of data source, econometric models and various econometric techniques that I will use to study the determinants of household consumption in Uganda.

### 3.1. Research design

This study will use secondary data following a quantitative research design to identify, analyze and describe the effects of income, savings and government expenditure on household consumption expenditure levels in Uganda.

### 3.2 Data source

The researcher will use secondary time series data collected from World Bank indicators, statistical abstracts, ministry of finance and economic development, budget speeches and background budget. The secondary data set will consist of government spending, income and saving observations (1).

### 3.3 Econometric model

#### Model specification

Household consumption expenditure =  $f(\text{GDP, savings, inflation, government spending})$ .

This can be expressed in the linear equation as:

$$C_t = \alpha_0 + \alpha_1 Y_t - \alpha_2 S_t + \alpha_3 G_t + \varepsilon_t \text{ where;}$$

$C_t$  = household consumption expenditure

$Y_t$  = public debt

$S_t$  = savings

$G_t$  = government expenditure

$\alpha_{0,1,2,3}$  = the regression coefficients will represent the relationships between the dependent variable and each of the independent variables in the model.

$\varepsilon_t$  = the error term is also called the stochastic error term will be used to represent all the other variables that may not be directly represented in the model due to the scope that will influence the model.

### **3.4. Estimation Techniques**

#### **3.4.1. Descriptive analysis**

The time series data will be descriptively analyzed to ensure the normality of distribution. These include measures of central tendency (mean, median, and mode) and a measure of variability (standard deviation, skewness, and kurtosis).

#### **3.4.2 Correlation analysis**

Correlation tests will be carried out to determine if a statistical relationship exists between then the variables and can one variable be predicted from another.

#### **3.4.3 Diagnostic test**

As a preliminary test for time series data, stationary tests will be carried out using unit root tests. Further, to prove the authenticity of the model and credibility to the empirical results, the regression test, multicollinearity test, normality test, heteroscedasticity test and autocorrelation tests will be carried out to ensure the stability of the model over time.

### **3.5 Limitations of the study**

Due to the nature of secondary data adapted in the investigation, some factors which would have been important for the study were left out such as questionnaires on the demographic variables such as age and gender that would be used to determine the differences in demographic trend in regards to household consumption expenditure in Uganda.

## CHAPTER FOUR

### PRESENTATION, ANALYSIS AND DISCUSSION OF FINDINGS

#### 4.0 Introduction.

This chapter presents and discusses the empirical results of the study on the factors that determine house hold consumption in the Ugandan economy. It gives the descriptive statistics, unit root tests, correlation and regression results, the heteroscedasticity test, autocorrelation test and their interpretations.

#### 4.1 Descriptive statistics.

The descriptive statistics of the variables employed in this study were presented the table below.

The table presents the mean, the median and the maximum values of both the dependent and the independent variables as below.

*Table 3 showing the descriptive statistics in the model variables*

| Variable    | obs | Mean     | MIN        | Max      |
|-------------|-----|----------|------------|----------|
| Consumption | 30  | 109.5251 | 102.562    | 115.9374 |
| Savings     | 30  | 12.02866 | -0.7661753 | 23.54522 |
| Public debt | 30  | 10.22001 | 1.319749   | 51.03291 |
| Expenditure | 30  | 3.394917 | 2.977267   | 3.822859 |

The mean consumption expenditure is approximately 109.5251, the mean savings, public debt and expenditure are 12.0866%,10.2201% and 3.394917% respectively, the table also gives the minimum and maximum values.

#### 4.2 Unit root tests.

The data variables are examined to ascertain whether the data is stationary or non-stationary in order to avoid spurious results. This was done using the Augmented Dickey Fuller Tests of the variables at their levels, first difference and second difference.

The unit root results are presented as follows:

##### 4.2.1 Testing for Stationarity for Consumption

Ho: Consumption is non-stationary

Ha: Consumption is stationary

From the original form of dickey fuller test, consumption was non-stationary and therefore it was differenced once to make them stationary as represented by the table below.

**Table 4 ADF TEST RESULTS**

| Interpolated Dickey Fuller                    |                |             |             |              |
|-----------------------------------------------|----------------|-------------|-------------|--------------|
|                                               | Test statistic | 1% critical | 5% critical | 10% critical |
| Z(t)                                          | -8.440         | -3.743      | -2.997      | -2.629       |
| Mackinnon approximate p-value for Z(t)= 0.000 |                |             |             |              |

It is therefore concluded that consumption is stationary since the ADF test statistics (8.440) > all the critical values.

Looking at the Mackinnon approximate p-value for Z(t)= 0.000, since the p-value (0.000 is less than the alpha value (0.05), we reject the null hypothesis in favor of the alternative hypothesis and conclude that consumption is stationary at 5% significance level.

#### **4.2.2 Testing for stationary for savings**

Ho: savings are non-stationary.

Ha: savings are stationary.

From the original form of dickey fuller test, Savings are non-stationary and therefore it was differenced once to make it stationary as represented by the table below.

**Table 5 ADF TEST RESULTS**

| Interpolated Dickey Fuller                    |                |             |             |              |
|-----------------------------------------------|----------------|-------------|-------------|--------------|
|                                               | Test statistic | 1% critical | 5% critical | 10% critical |
| Z(t)                                          | -7.034         | -3.743      | -2.997      | -2.629       |
| Mackinnon approximate p-value for Z(t)= 0.000 |                |             |             |              |

It is therefore concluded that savings are stationary since the ADF test statistics (7.034) > all the critical values.

Looking at the Mackinnon approximate p-value for  $Z(t) = 0.000$ , since the p-value (0.000 is less than the alpha value (0.05), we reject the null hypothesis in favor of the alternative hypothesis and conclude that savings are stationary at 5% significance level.

#### 4.2.3 Testing for stationary for public debt

Ho: Public debt is non-stationary

Ha: Public debt is stationary

From the original form of dickey fuller test, Public debt was non-stationary and therefore it was differenced once to make it stationary as represented by the table below.\

*Table 6 ADF TEST RESULTS*

| Interpolated Dickey Fuller                       |                |             |             |              |
|--------------------------------------------------|----------------|-------------|-------------|--------------|
|                                                  | Test statistic | 1% critical | 5% critical | 10% critical |
| Z(t)                                             | -7.627         | -3.743      | -2.997      | -2.629       |
| Mackinnon approximate p-value for $Z(t) = 0.000$ |                |             |             |              |

It is therefore concluded that Public debt is stationary since the ADF test statistics (-7.627) > all the critical values.

Looking at the Mackinnon approximate p-value for  $Z(t) = 0.000$ , since the p-value (0.000 is less than the alpha value (0.05), we reject the null hypothesis in favor of the alternative hypothesis and conclude that public debt is stationary at 5% significance level.

#### 4.2.4 Testing for stationary for Expenditure

Ho: Expenditure is non-stationary

Ha: Expenditure is stationary

From the original form of dickey fuller test, Expenditure is non-stationary and therefore it was differenced once to make it stationary as represented by the table below.

**Table 7 ADF TEST RESULTS**

| Interpolated Dickey Fuller                    |                |             |             |              |
|-----------------------------------------------|----------------|-------------|-------------|--------------|
|                                               | Test statistic | 1% critical | 5% critical | 10% critical |
| Z(t)                                          | -13.036        | -3.743      | -2.997      | -2.629       |
| Mackinnon approximate p-value for Z(t)= 0.000 |                |             |             |              |

It is therefore concluded that expenditure is stationary since the ADF test statistics (-13.036) > all the critical values.

Looking at the Mackinnon approximate p-value for Z(t)= 0.000, since the p-value (0.000 is less than the alpha value (0.05), we reject the null hypothesis in favor of the alternative hypothesis and conclude that expenditure is stationary at 5% significance level.

#### **4.3 Correlation analysis.**

Basically, the insight in the correlation analysis is to ascertain how the different variables are related to one another and more specifically the dependent and explanatory variables. Correlation results are as below.

**Table 8 THE CORRELATION TEST BETWEEN THE INDEPENDENT AND THE DEPENDENT VARIABLES**

| Variables   | Consumption | Savings | Public Debt | Expenditure |
|-------------|-------------|---------|-------------|-------------|
| Consumption | 1.0000      |         |             |             |
| Savings     | -0.7321     | 1.0000  |             |             |
| Public Debt | 0.2787      | -0.6686 | 1.0000      |             |
| Expenditure | -0.4830     | 0.8859  | -0.6825     | 1.0000      |

There is a high negative correlation between savings and consumption, expenditure is also found to have a negative correlation with consumption, however public debt was found to have a positive correlation with consumption expenditure.

#### 4.4 Regression analysis

The regression analysis below shows the effects between the dependent variables and the independent variables, it also portrays the R-square value and also the level of significance of the variables.

**Table 9 SHOWING REGRESSION RESULTS**

| Consumption | Coef.    | Std. Err. | t-value       | p-value | [95% Conf. Interval] |           |
|-------------|----------|-----------|---------------|---------|----------------------|-----------|
| Savings     | -0.06487 | 0.0146504 | -4.43         | 0.000   | -0.0951              | -0.034633 |
| Public debt | -0.25455 | 0.0061651 | -4.13         | 0.000   | -0.0381              | -0.012731 |
| Expenditure | -0.04399 | 0.1147524 | -0.38         | 0.705   | -0.2808              | 0.192841  |
| Constant    | 4.945943 | 0.1231956 | 40.15         | 0.000   | 4.69168              | 5.200206  |
| R-squared   |          | 0.6784    | Number of obs |         | 28                   |           |
| F-test      |          | 16.87     | Prob > F      |         | 0.0000               |           |

#### Interpretations

The R-square value (0.6784) implies that 67.8% of the variations in the consumption expenditure can be explained by the variations in savings, public debt, and expenditure hence a good fit.

When savings, public debt and expenditure are zero, the household consumption rate is approximately 4.945943% and this is statistically significant at 5% level of significance since the value of the t-value is greater than 2 in absolute terms.

A one percent increase in the savings levels on average leads to a 0.06487% decrease in the household consumption other factors held constant and this is significant at 5% level of significance based on the value of the t-value.

A one percent increase in the public debt on average leads to a 0.25455% decrease in the consumption expenditure and is statistically significant at 5% level of significance since the t-value is greater than 2 in absolute terms other factors being held constant.

A one percent increase in expenditure on average leads to a 0.04399% decrease in the house hold and is not statistically significant at 5% level of significance other factors being held constant.

#### **4.5 Testing for Heteroscedasticity**

Ho. Homoscedasticity

Ha. Heteroscedasticity

*Table 10 Heteroscedasticity results*

| Prob>Chi2 | Chi 2 (1) |
|-----------|-----------|
| 0.9534    | 0.00      |

Since the probability value (0.9534) is greater than alpha value (0.05), we fail to reject the null hypothesis and conclude that there is homoscedasticity at 5% significance level

#### **4.6 Test for Multi collinearity**

Ho. There is no Multicollinearity

Ha. There is Multicollinearity

*Table 11 Multi collinearity results*

| Variable    | VIF  | 1/VIF    |
|-------------|------|----------|
| Expenditure | 5.43 | 0.184018 |
| Savings     | 3.73 | 0.267987 |
| Public debt | 2.41 | 0.414296 |
| Mean VIF    | 3.86 |          |

Since the mean VIF= 3.86 and is less than 10, there is no multi collinearity problem within the model.

#### **4.7 Testing for autocorrelation**

Ho: No autocorrelation

Ho: there is autocorrelation

**Table 12 Auto correlation test results**

|                                |             |                     |
|--------------------------------|-------------|---------------------|
| <b>Breusch-Godfrey LM test</b> | <b>Df3.</b> | <b>Prob&gt;chi2</b> |
| <b>Chi2</b>                    |             |                     |
| <b>0.083</b>                   | 1           | 0.7727              |

The observed probability chi-square value (0.7727) is greater than the 0.05. As a result, the null hypothesis of no autocorrelation is not rejected and we conclude that there is no autocorrelation among the variables.

#### **4.8 Testing for normality**

Ho: the data follows a uniform distribution.

Ha: the data does not follow a uniform distribution.

**Table 13 test for normality results**

| variable | obs | w       | v     | z     | Prob>Z  |
|----------|-----|---------|-------|-------|---------|
| Hillary  | 28  | 0.96281 | 1.123 | 0.239 | 0.40557 |

## **CHAPTER FIVE**

### **SUMMARY, CONCLUSIONS AND RECOMMENDATIONS.**

#### **5.0 INTRODUCTION**

This chapter presents the summary of the research study, conclusions, policy recommendations and areas for further research.

#### **5.1 summary of findings**

This study examined the determinants of House hold consumption expenditure in Uganda using time series data for the period 1990 to 2020. The macroeconomic variables that were used in this case included public debt, Consumption, government expenditure and domestic savings. The study conducted a unit root test using Augmented Dickey Fuller Test to check for the presence of a unit root aimed at avoiding spurious regression results. The tests showed that all variables had a unit root and were differenced to induce stationarity. Normality tests were further carried out to ascertain whether the data for the variables was normally distributed. However empirical results showed that the data was normally distributed. Diagnostic tests such as multi-collinearity, heteroscedasticity, auto correlation was carried out for the study variables used in the research paper. The model estimates showed that consumption, savings, expenditure and public debt exerts a negative but very statistically significant impact on household consumption expenditure in both the short run and long run. However, public expenditure had a negative but statistically insignificant relationship on the household consumption expenditure.

#### **5.2 Conclusion**

The study concludes that the Savings (income) has a positive impact on the House hold consumption, this implies that an increase in the gross domestic income leads to an increase in the gross domestic savings both in the short run and the long run. This is however not statistically significant at 5% level of significance.

The study also on the other hand concludes that public debt and expenditure on the other hand has a negative impact on the household consumption expenditure, this implies that as public debt and expenditure increases, the household consumption on the other hand decreases both in the short run and the long run and they are statistically significant at 5% level of significance.

#### **5.3. Recommendations**

Based on the findings and conclusions of the study on the determinants of households and private consumption expenditure, the following recommendations can be made.

**Promoting financial education and saving habits:**

Implement educational programs and initiatives to enhance financial literacy and promote moderate savings habits among households. Encouraging individuals to save a portion of their income can lead increased gross savings, which in turn stimulates private consumption expenditure and contributes to economic growth.

**Address population dynamics.**

Consider population growth rates and their impact on private consumption expenditure when formulation economic strategies. Develop policies that address the needs and demands of a growing population to support consumer spending and economic development. This may involve infrastructure investment, employment generation, and social welfare programs.

**Foster a supportive policy environment.**

Create an enabling policy environment that encourages entrepreneurship, and investment. This can lead to increased productivity, income growth, and ultimately, higher private consumption expenditure. Policy measures such as reducing bureaucratic hurdles, providing incentives for business growth and supporting research and development can help foster economic dynamism and consumer confidence.

**Continuously monitor and evaluate consumer behavior.**

Regularly collect and analyze data on households' private consumption expenditure, gross savings, inflation and other relevant variables. This ongoing monitoring and evaluation can provide valuable insights into changing consumer behavior and policymakers make informed decisions and adjustments to economic policies as needed.

Fostering international cooperation and coordination to address global economic challenges. Collaborate with other counties to tackle issues such as cross-border capital flows, international trade imbalances, and exchange rate fluctuations, which can have implications for households' consumption patterns. Cooperation in areas such as monetary policy coordination and trade agreements can contribute to stable and sustainable private consumption growth.

#### **5.4 Areas for further research**

This study has investigated the determinants of house hold consumption expenditure in Uganda for the period 1990 to 2020 using annual data from World Bank. The paper recommends that further research should be done using cross sectional data in order to capture the household consumption expenditure determinants. This is because the explanatory variables examined in this study are not the only determinants of Uganda's household consumption expenditure. There are many other determinants of house hold consumption expenditure that have not been included in the study.

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